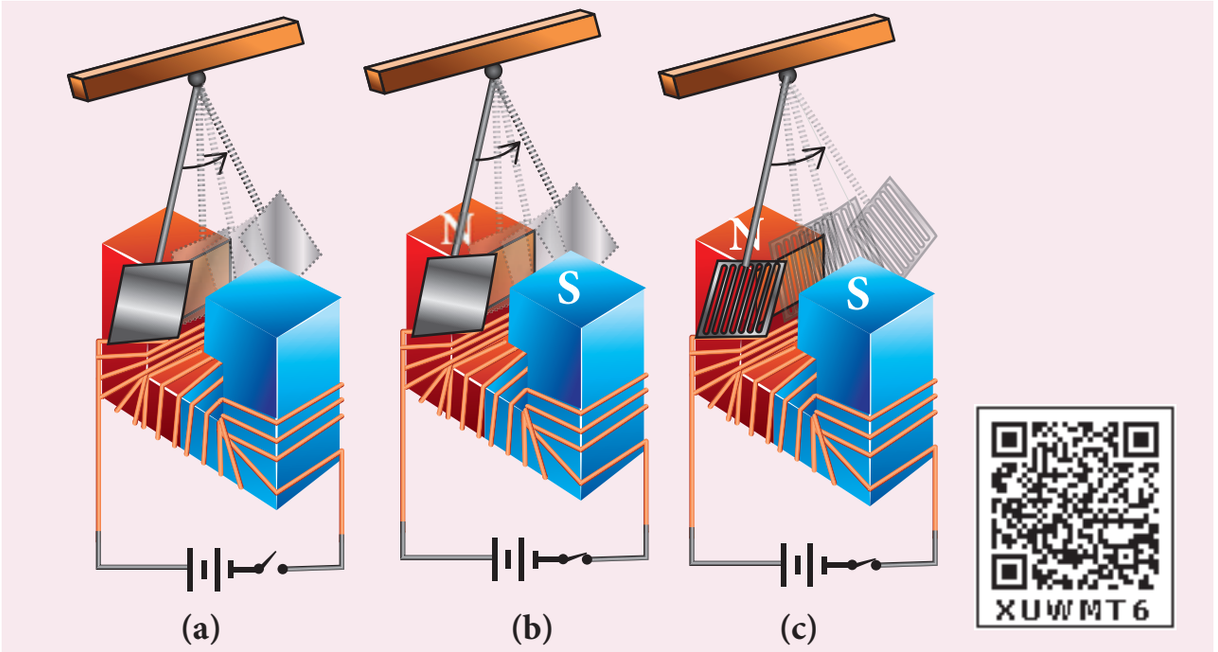


Figure 4.12 Demonstration of eddy currents

However if some slots are cut in the disc as shown in Figure 4.12(c), the eddy currents are reduced. The pendulum now will execute several oscillations before coming to rest. This clearly demonstrates the production of eddy current in the disc of the pendulum.

Drawbacks of Eddy currents

When eddy currents flow in the conductor, a large amount of energy is dissipated in the form of heat. The energy loss due to the flow of eddy current is inevitable but it can be reduced to a greater extent with suitable measures.

The design of transformer core and electric motor armature is crucial in order to minimise the eddy current loss. To reduce these losses, the core of the transformer is made up of thin laminas insulated from one another (Figure 4.13 (a)) while for electric motor the winding is made up of a group of wires insulated from one another (Figure 4.13 (b)). The insulation used does not allow huge eddy currents to flow and hence losses are minimized.

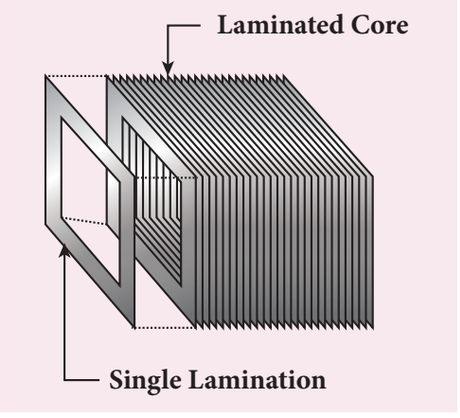


Figure 4.13 (a) Insulated laminas of the core of a transformer

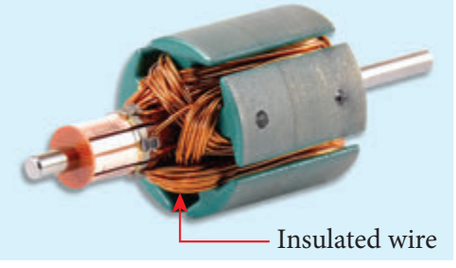


Figure 4.13 (b) Insulated winding of an electric motor

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